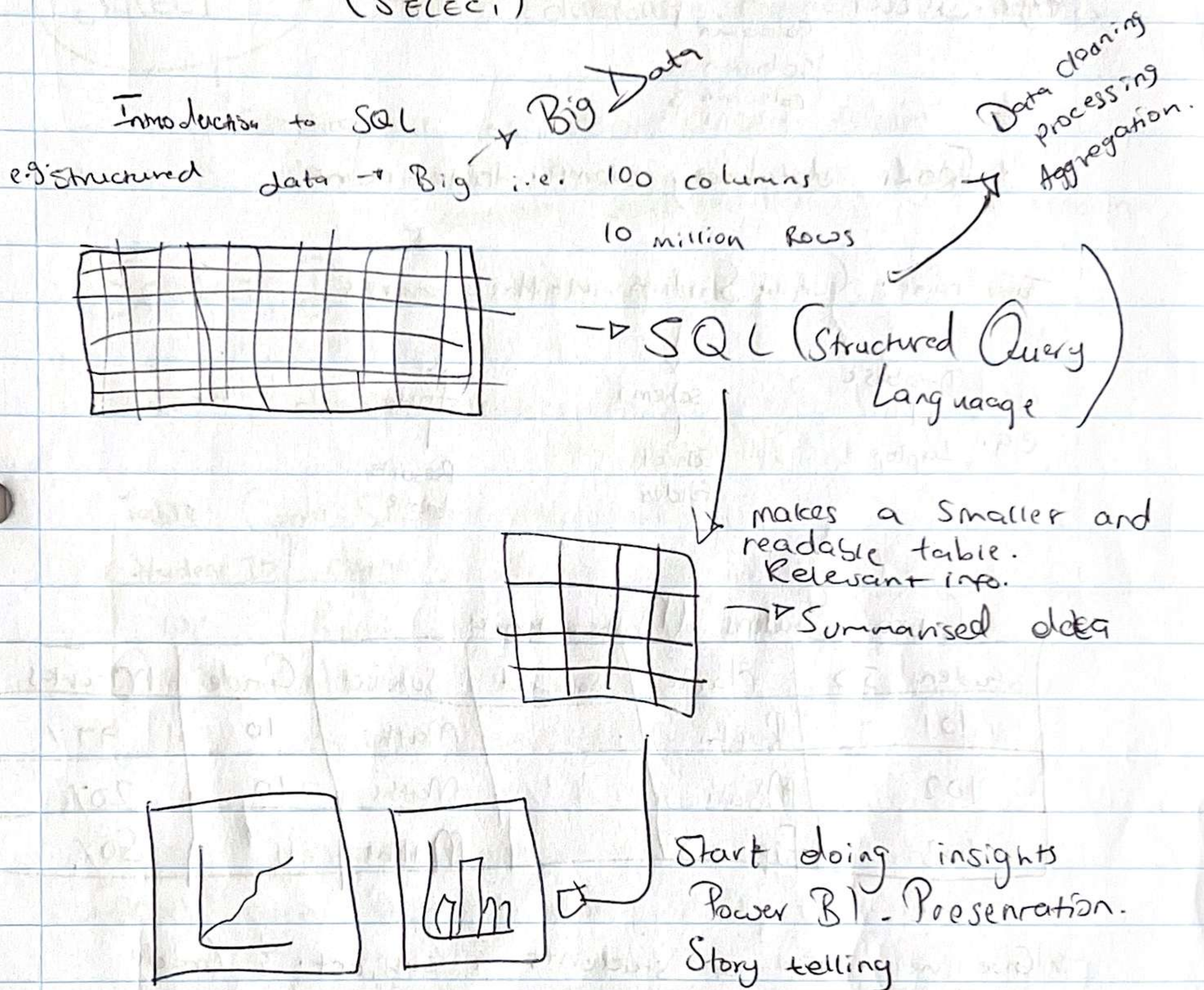


Lesson 1: SQL fundamentals (SELECT)



SQL Syntax

↳ The way we write SQL

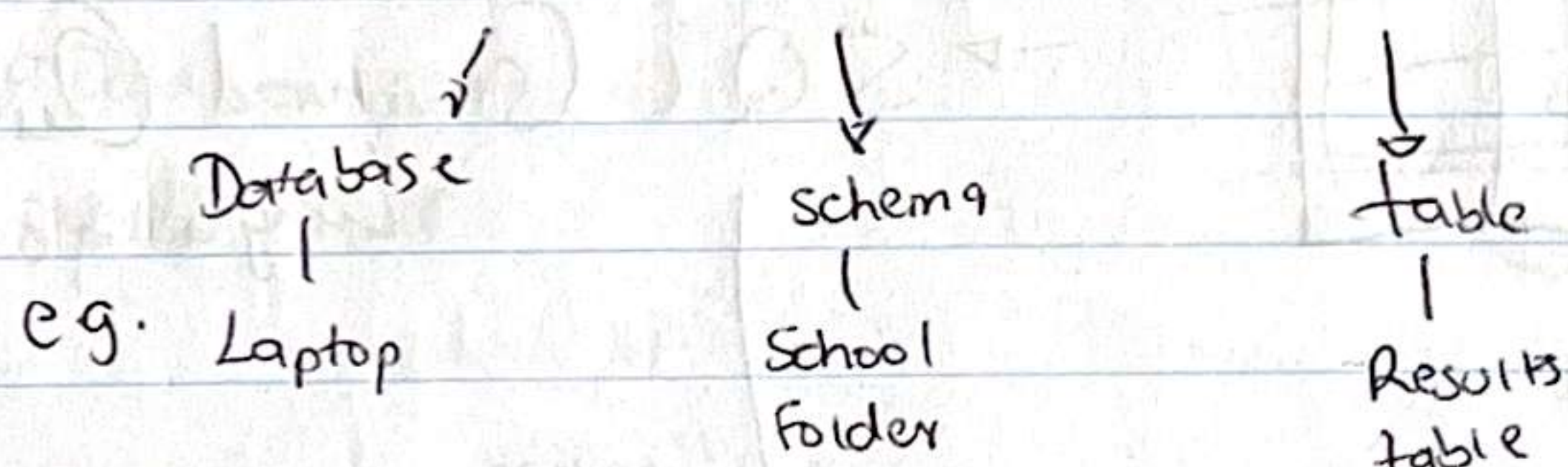
- **SELECT** statement / clause
 - Allows to specify diff columns to be used from a specific table.
- **FROM** statement (DATABASE . SCHEMA . TABLE)
 - Allow you to specify which table you are selecting from.
 - Specify the name of the database, schema, table-name

eg. Select

column 1
column 2
column 3

from database . schema . table - name

Table name: Curro Students . Maths - marks



Curro . Students . Maths - marks

Student ID	Name	Subject	Subject	Grade	Marks
101	Rafhi		Maths	10	99%.
102	Mbali		Maths	10	20%.
103	Fifi		Maths	11	50%.

- & Give me a list of students & subject & Grade

Select

name
Subject
Grade

from

Curro . Students . Maths - marks;

Name	Subject	Grade %
Rafhi	Maths	99
Mbali	Maths	20
Fifi	Maths	50

Select * from database . schema . table - name ;

↓ Selecting all the columns in the table.

↗ selecting different values in the column

Select Distinct Column name
from database . schema . table - name ;

Table: Camo . Students . maths - marks

Student ID	Name	Subject	Grade	Marks
101	Raphi	Maths	10	99
102	Mbali	Maths	10	20
103	fifi	Maths	11	50

Select Distinct grade
from camo . Students . maths - marks .

↓

Grade
10
11

↗ Shows only the diff values in the columns.

Select Statement

Select

↳ Used to choose which columns you want to see on final product

Select *

↳ Returns every column in the table

Select Distinct

↳ Returns unique values from the table.

Select

column1, column2, etc.

↳ Returns multiple columns from the table

↳ Use commas between column names except last column name.

Lesson 2: SQL Fundamentals

- Filtering Statements

WHERE Statement

↳ Used to specify conditions in order to filter rows in the table.

- Filtering is ~~for~~ ^{the} Rows

NB: Select statement is for filtering columns.

SQL Syntax:

Select

column 1,
column 2,
column 3,
column 4

↳ No comma at the last column.

From

database.schema.table;

Where

condition;

e.g. Table Curro. Student. marks

student ID	Name	Subject	marks
01	Rophi	Maths	99
102	Mbali	Maths	22
103	Fifi	Geo	50
104	Tumelo	Bio	40

i.e. Show names of students with marks less than 50.

Select

name,
marks

From

curro.students.marks

Where

marks < 50;

↳ Filtered the rows

Name	Marks
Mbali	22
Tumelo	40

SQL Comparison Operators

→ used to filter in order to build the condition for filtering

Operator	Description
=	Equal to
>	greater than
>=	greater than or equal to
<	Less than
<=	less than or equal to
<>	Not equal to
!=	Not equal to

→ Used within the WHERE clause / statement in order to filter data.

Student ID	Name	Subject	Marks
101	Rofhi	Marks	99
102	Mbali	Maths	22
103	fifi	Geo	50
104	Tumelo	Bio	40

SELECT *
FROM curro.student.marks
WHERE Subject = 'Geo'

→ All columns
→ Single quotes are used because Geo is a word not a number

Ans :

Student ID	Name	Subject	Marks
103	fifi	Geo	50

Text Filtering.

- Text inside the table, use single quotes ' '
- Naming a column, no quotes
- Numbers, no quotes

Lesson 3

SQL Logical Operators

* Helps us to build specific logic

used in WHERE statements.

AND - Helps/Allows us to include both conditions ^{multiple}

OR - One condition be met

IN - Specify values you want

NOT - Gives opposite results.

BETWEEN - Inclusive

AND operator

↳ A statement that helps to join multiple conditions

↳ Both conditions must be met.

e.g. Calling students who are doing both maths & science.

Select Column 1,
Column 2,
Column 3

From database.Schema.table

Where condition 1 AND condition 2

Curro. Students - marks

Student-ID	Name	Subject	Marks %
101	Rofhi	Maths	99
102	Mbali	Maths	22
103	Lerato	Geo	50
104	Ally	Acc	40

* Name & Marks of students ^{50%} doing who got above 50% in maths

Select name,
marks

From curro.Students.marks

Where Subject = 'maths' AND Marks > 50

OR Statement

↳ Allow for specification of multiple conditions, but ^{at least} only 1 condition must be met.

Select column1
column2

from database.schema.table

Where condition1 OR condition2

e.g. Students doing maths or got 50% or more

Select *

From curro.students.marks

Where Subject = 'maths' OR marks >= 50

ID	Name	Subject	Marks
101	Rofhi	maths	99
103	Lerato	Geo	50
102	Mbuthi	maths	22

BETWEEN Operator

↳ Allows us to filter data using a range of values. An inclusive statement/operator

Select column1,
column2,
column3

From database.schema.table

Where condition
column-name BETWEEN minimum AND Maximum

Use curro.Students.Marks table.

i.e. Write an SQL query to get name & marks of students who obtain marks between 0-30.

Select name, marks, subject

From curro.student.marks

Where marks BETWEEN 0 AND 30;

Name	Subject	Marks
Mbuthi	maths	22

ie. Marks between 50 & 100.

Select *

from auto.students.marks

Where marks Between 50 AND 100;

Student-ID	Name	Subject	Marks
101	Rofhi	Maths	99
103	Lerato	Geo	50

IN operator

↳ Allows us to specify specific values in a column that we want to include.

↳ The values not included will be excluded from the final outcome.

SELECT column 1
column 2

FROM database.schema.table

WHERE column-name IN (value 1, value 2, value 3)

e.g. Shoprite.online.Sales

customer-ID	customer name	city
111	Rofhi	JHB
112	mbali	CPT
113	Lerato	JHB
114	Thaboo	PTA

⇒ Show customers in JHB & Pretoria

Select customer name, city

from shoprite.online.sales

Where city IN ('JHB', 'PTA');